

Key Observations Report

Exploratory Data Analysis (EDA) — Project 2

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1. Problem Statement

The core question being analyzed is: What patterns and trends exist in e-commerce order data? The analysis focuses on understanding pricing behavior, spending distributions, outliers, relationships between key numerical variables, and customer acquisition sources across 1,200 transactions.

2. Methodology

The dataset contains 1,200 order records. The following analytical steps were performed:

- Descriptive statistics calculated for all numerical columns: Quantity, UnitPrice, ItemsInCart, and TotalPrice
- Metrics computed: Count, Mean, Median, Minimum, Maximum, Q1, Q3, and IQR
- Outlier detection applied to TotalPrice using the IQR method
- Correlation analysis conducted between three key variable pairs
- Referral source analysis using COUNTIF to count orders by customer acquisition channel

3. Descriptive Statistics Summary

Metric	Quantity	Unit Price	Items In Cart	Total Price
Count	1200	1200	1200	1200
Mean	2.95	\$356.41	5.49	\$1,053.97
Median	3	\$364.21	5	\$823.62
Minimum	1	\$11.39	1	\$11.39
Quartile 1	2	\$186.06	4	\$410.52
Quartile 3	4	\$521.57	7	\$1,578.48
Maximum	5	\$699.93	10	\$3,456.40
IQR	2	\$335.51	3	\$1,167.95
Upper Bound	-	-	-	\$3,330.41
Total Outliers	-	-	-	8

Correlation Analysis

Pearson correlation was calculated between the following variable pairs:

Variable Pair	Correlation	Interpretation
UnitPrice vs. TotalPrice	0.717	Strong positive correlation
ItemsInCart vs. TotalPrice	0.393	Moderate positive correlation
UnitPrice vs. Quantity	0.015	Almost zero correlation

Referral Source Analysis

The following table shows the number of orders acquired through each customer referral channel:

Referral Source	Order Count	Percentage
Instagram	259	21.6%
Email	250	20.8%
Google	241	20.1%
Facebook	228	19.0%
Referral	222	18.5%

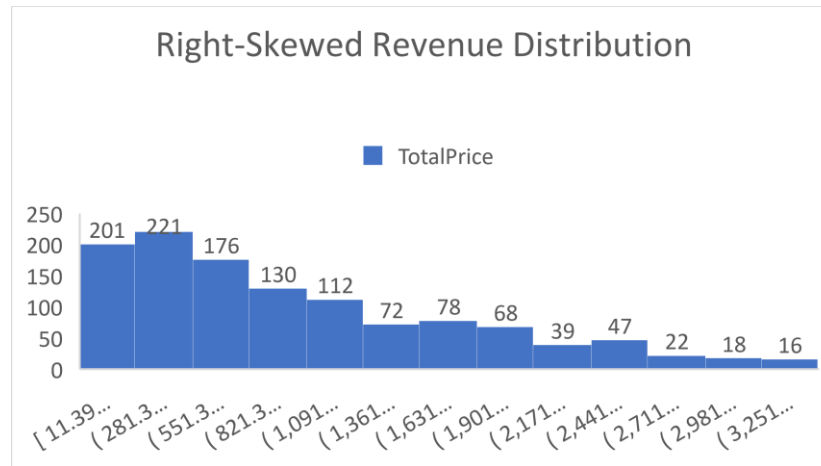
4. Key Findings

Finding 1 — Average order value is higher than the typical order value

The mean TotalPrice is \$1,053.97 while the median is \$823.62. Since the mean is noticeably higher than the median, the data is right-skewed — meaning a few large orders are pulling the average up. The median is the more accurate representation of a typical order.

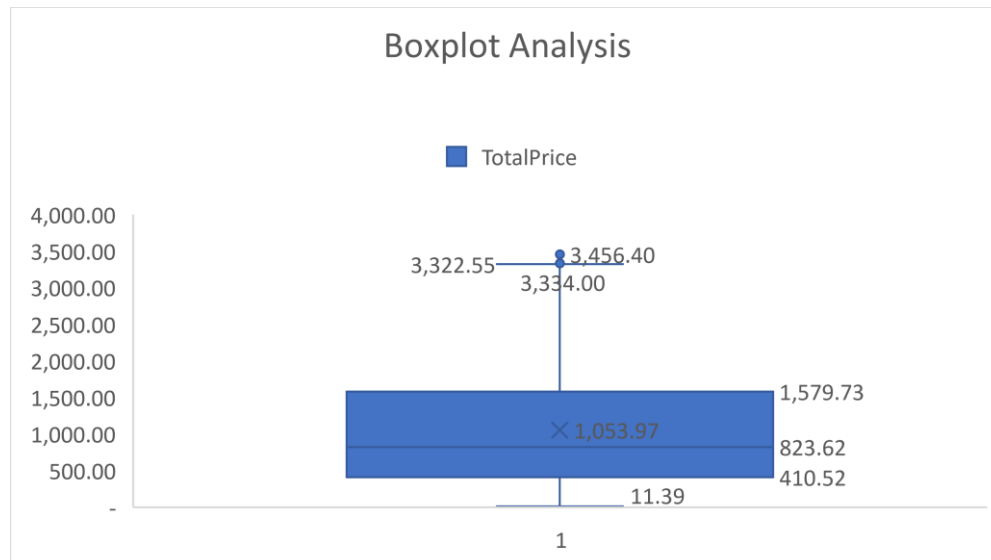
Finding 2 — TotalPrice has a wide spread

TotalPrice ranges from a minimum of \$11.39 to a maximum of \$3,456.40, with a large IQR of \$1,167.95 (Q1: \$410.52, Q3: \$1,578.48). This shows customers have very different spending levels — some buy very cheaply while others spend significantly more.



Finding 3 — 8 High-Value Outliers were identified

Using the IQR method, the upper bound was calculated as \$3,330.41. A total of 8 orders exceeded this threshold and were flagged as High Value. These are genuine high-spending customers worth further attention.



Box plot of TotalPrice showing distribution from \$11.39 to \$3,456.40. The dot above the upper whisker represents high-value outlier orders exceeding the IQR upper bound of \$3,330.41, consistent with the 8 outliers identified in the analysis.

Finding 4 — UnitPrice is the strongest driver of TotalPrice

The correlation between UnitPrice and TotalPrice is 0.717 (strong positive). This means the price of the product is the biggest factor in how much a customer ends up spending. Higher-priced products directly lead to higher total order values.

Finding 5 — Cart size has a moderate influence on spending

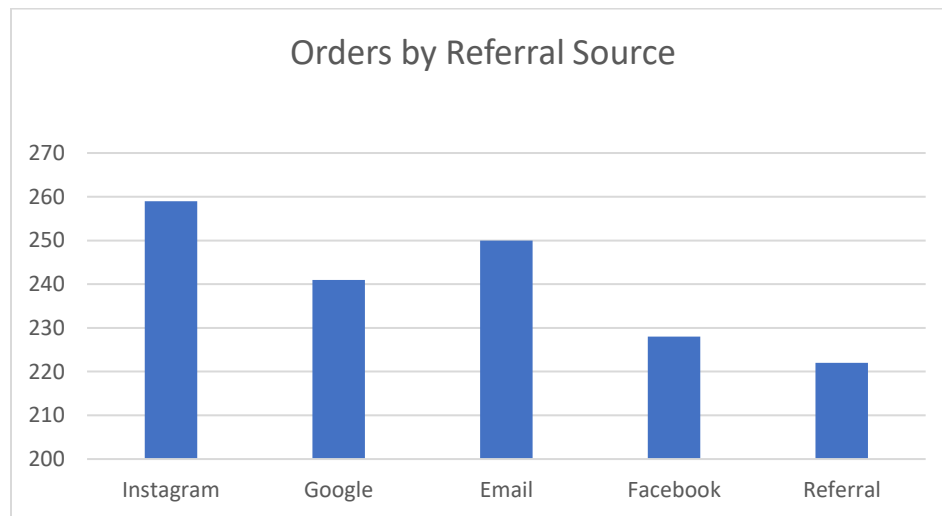
ItemsInCart vs TotalPrice shows a moderate positive correlation of 0.393. Customers with more items in their cart do spend more, but this relationship is not strong enough to be the primary revenue driver.

Finding 6 — Quantity ordered has almost no relationship with UnitPrice

The correlation between UnitPrice and Quantity is nearly zero (0.015). Customers buy expensive and cheap products in similar quantities — there is no bulk-buying pattern observed for cheaper items.

Finding 7 — Instagram is the top customer acquisition channel

Instagram drives the highest number of orders at 259, slightly ahead of Email (250) and Google (241). Facebook (228) and Referral (222) trail behind. All five channels are fairly balanced, but Instagram holds a slight edge as the leading source of customer acquisition.



5. Recommendations

1. **Use median (\$823.62) rather than mean for reporting typical order value** — the mean is distorted by outliers and gives a misleading picture.
2. **Focus on selling higher-priced products** — since UnitPrice is the strongest revenue driver ($r = 0.717$), promoting premium items will have the most impact on revenue.
3. **Investigate the 8 High-Value outlier orders** — these customers spent well above average and could be VIP buyers worth targeting with loyalty offers.
4. **Do not over-invest in cart-quantity strategies** — since cart size only moderately affects spend ($r = 0.393$), product pricing matters far more than adding more items to the cart.
5. **Prioritize Instagram as the primary marketing channel** — with the highest order count of 259, increasing investment in Instagram content and ads is likely to generate the best customer acquisition return.